

Domain Knowledge

Terminology

Tropical Cyclone

(what.) A tropical cyclone is the generic scientific term for a rotating, organised system of clouds and thunderstorms that forms over tropical/subtropical waters. It has closed low-level circulation and draws its energy primarily from the ocean.

Conditions

The necessary (but not always sufficient) conditions for a tropical cyclone (formalised by William Gray):

1. **Warm Ocean Water:** The sea surface temperature must be $\geq 26.5^{\circ}\text{C}$ through a sufficient depth $\geq 50\text{m}$ → Supplies latent heat that powers the storms convection
2. **Atmospheric Instability:** The atmosphere must cool rapidly enough with height → Allows moist air parcels to rise freely and produce deep thunderstorms
3. **Moist Mid-Troposphere:** Relatively high relative humidity in the lower-to-middle troposphere $\sim 5\text{km}/700\text{-}500\text{ hPa}$ → Dry air at these levels suppresses sustained convection
4. **Sufficient Coriolis Force:** The system must be at least $\sim 500\text{ km}$ (5° of latitude) from the equator → Otherwise without enough planetary rotation, closed circulation cannot organise and be maintained
5. **Pre-existing low-level disturbance:** Some seed of vorticity and convergence must already exist, e.g. African easterly wave, monsoon trough, old frontal boundary, or convective cluster → Tropical cyclones do not form spontaneously from a completely calm environment
6. **Low vertical wind shear:** The change in wind speed and direction between the lower and upper troposphere should generally be less than $\sim 10\text{ m s}^{-1}$ ($\approx 20\text{ kt}$) → Strong shear tilts and disrupts the developing vortex

Names

Once the above conditions are met and the system intensifies, names depend on location and intensity:

Name	Ocean basin / region	Wind threshold for the name	Notes on intensity scales
Hurricane	North Atlantic, Caribbean, Gulf of Mexico, eastern & central North Pacific (east of the Date Line)	≥ 64 kt (74 mph / 119 km/h)	Uses the Saffir-Simpson Hurricane Wind Scale (Categories 1-5)
Typhoon	Northwestern Pacific (west of the Date Line)	≥ 64 kt (same physical threshold)	“Super typhoon” is often used for the most intense systems (≥ ~130-150 kt depending on agency)
Cyclone (or severe tropical cyclone / very severe cyclonic storm)	South Pacific, Indian Ocean (including Bay of Bengal and Arabian Sea)	≥ 64 kt (same physical threshold)	Local scales vary (e.g. Australian categories, IMD categories in the North Indian Ocean)

A storm with identical structure and intensity would be called a hurricane if it formed in the Atlantic, a typhoon if it formed in the western North Pacific, and a cyclone (or severe tropical cyclone) if it formed in the Indian Ocean or South Pacific

Regional Differences

- **Peak seasons**
 - Atlantic hurricanes: June-November (peak mid-September)
 - Northwest Pacific typhoons: year-round, with peaks typically May-October
 - Southern Hemisphere cyclones: November-April (peak January-March)

- North Indian Ocean: bimodal (pre- and post-monsoon peaks)
- **How often the conditions are met**

The western North Pacific has the most favourable large-scale environment overall and produces the most storms. Near-equatorial formation is also more common there than in the Atlantic because of differences in low-level vorticity and shear patterns
- **Intensity classification details**

Different agencies use slightly different sustained-wind averaging periods (1-minute vs 10-minute) and different category boundaries, but the physical 64 kt threshold for “hurricane-force” winds is consistent
- **Local environmental nuances**

Some basins experience more frequent dry-air intrusions, different shear regimes, or shallower warm-water layers, which affect how often RI occurs or how intense storms become

These are environmental differences, not differences in the fundamental formation requirements